VIJAY YADAV

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EDUCATION

Ph.D. Physics — 12/2021 – expected 04/2027
Indian Institute of Technology Jodhpur — Rajasthan, India

- Thesis title: Data-Driven Surrogate Models for Phase-Ordering Kinetics and Charge Regulation Effects.
- PhD advisor: Prabhat K. Jaiswal

M.Sc. Physics — 08/2018 – 06/2020
National Institute of Technology Tiruchirappalli — Tamil Nadu, India

B.Sc. (Hons.) Physics — 07/2015 – 06/2018
University of Delhi — New Delhi, India

RESEARCH INTERESTS

My research lies at the intersection of soft matter physics, molecular simulation, and scientific machine learning. I am interested in developing physics-guided, data-driven surrogate models that accelerate molecular and continuum simulations, enabling access to longer timescales and the exploration of large parameter spaces. I also use hybrid molecular dynamics/Monte Carlo (MD/MC) methods to study charge regulation effects in soft matter. Recently, I developed charge regulation machine learning (CR-ML), a hybrid approach that accelerates the charge-update step of these simulations by more than an order of magnitude.

Keywords: Data-Driven Surrogates, Dynamical Systems, Charge Regulation, Molecular Dynamics, Monte Carlo

PUBLICATIONS

Published

1. V. Yadav, P. K. Jaiswal, R. Podgornik, and S. Kumari, Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte, *Macromolecules* **59**, **11**, 6398-6409 (2026).
2. V. Yadav, M. Priya, M. D. Shrimali, and P. K. Jaiswal, Graph Neural Network for Prediction of Phase-Ordering Kinetics, *Chaos* **35**, 061101 (2025).
3. S. Chauhan, S. Mandal, V. Yadav, P. K. Jaiswal, M. Priya, and M. D. Shrimali, Machine Learning Based Prediction of Phase Ordering Dynamics, *Chaos* **33**, 061103 (2023).

In Review

1. V. Yadav, P. Pandey, M. Priya, M. D. Shrimali, and P. K. Jaiswal, Physics-Constrained Neural Surrogate for Domain Growth Prediction in Systems with Conserved Kinetics. arXiv:2606.26128
2. V. Yadav, S. Kumari, and P. K. Jaiswal, Machine Learning Accelerated Simulation of Charge Regulation Effects.

CONFERENCES & PRESENTATIONS

07/2026: Flash Talk: Data-Driven Modelling of Phase-Ordering Kinetics – Symposium on Complex Dynamics in Chemistry, Physics, and Biology (CDCPB 2026), IIT Jodhpur, Rajasthan, India.
03/2026: Oral Talk: Machine Learning and GPU-Accelerated Modeling of Phase-Ordering Kinetics – APS Global Physics Summit 2026, Denver, Colorado, USA. [Virtual].
03/2026: Poster Presentation: Charge Regulation Effects on Polyelectrolyte-Mediated Nanoparticle Interactions – APS Global Physics Summit 2026, Denver, Colorado, USA. [Virtual].
12/2025: Poster Presentation: Graph Neural Network for Phase-Ordering Kinetics – International Conference on Complex Fluids, IISc, Bangalore, India.
05/2023: Poster Presentation: Study of Hessian Matrix Eigenvalues of the Glassy System Before and After Yield Stress and Its Prediction Using Random Matrix Theory – CECAM Workshop on Mesoscale Modelling of Driven Disordered Materials: From Glasses to Active Matter, EPFL, Lausanne, Switzerland.

TECHNICAL SKILLS

Programming Languages

Python | Mojo

Deep Learning

Graph Neural Networks | Neural Operators | PyTorch | PyTorch Geometric | JAX/FLAX

Simulation

Molecular Dynamics | Monte Carlo | Finite Difference Method | LAMMPS

Additional Tools

HPC/SLURM | Git | OVITO | Matplotlib | LaTeX

AWARDS & ACHIEVEMENTS

06/2026: Article *Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte* selected for the front cover of *Macromolecules*, Vol. 59, Issue 11.

05/2023: Institute travel grant to present at the CECAM workshop at EPFL, Lausanne, Switzerland.

TEACHING & MENTORSHIP

Teaching Assistant

Indian Institute of Technology Jodhpur — 2023 – 2025

- Delivered lectures and tutorials on Python programming, algorithms, numerical methods, and data visualization for B.S. and M.Sc. students, with the largest class exceeding 30 students.

B.S. and M.Sc. Student Mentor

Indian Institute of Technology Jodhpur — 2022 – 2026

- Mentored four M.Sc. and two B.S. students on their thesis research projects.

LANGUAGES

English — Fluent

Hindi — Mother Tongue

OPEN-SOURCE PROJECTS

Simpok — `pip install simpok`

A Python package for phase-field simulation of phase-ordering kinetics, with all compute kernels written in Mojo. It provides Time-Dependent Ginzburg–Landau (Model A), Cahn–Hilliard–Cook (Model B), and surface-directed spinodal decomposition solvers. Same code runs on CPU and a GPU backend (NVIDIA, AMD, Apple).